AI-Supported Personalization for Assessing Maker Learning in K–12 Classrooms

Vimala Telu

Department of Learning Sciences, Georgia State University

LT 8300: Design Research in Practice

Dr. Min Kyu Kim

March 1, 2026

Research Setting

This study focuses on K–12 classrooms that integrate maker education into their STEM instruction. These classrooms feature makerspaces, design challenges, and hands-on engineering tasks in which students build physical or digital artifacts. The student population is diverse in language background, academic readiness, and prior experience with hands-on learning. Teachers typically work with 20–30 students who progress through open-ended tasks at different paces.

During maker activities, students use materials, tools, and digital platforms to design, test, and revise their ideas. Maker education supports the development of participatory, creative, and design-based competencies, which are essential for students to engage meaningfully in hands-on learning (Kafai & Peppler, 2011). Teachers circulate to monitor progress, answer questions, and provide feedback. However, because students work simultaneously and independently, teachers cannot observe each student's reasoning or decision-making. As a result, much of the learning process, such as how students troubleshoot, iterate, or respond to setbacks, remains invisible unless students explain their thinking verbally.

This setting poses a practical challenge: teachers need reliable ways to understand how students approach problems, revise their designs, and respond to setbacks. These insights are essential for supporting learning, yet they are difficult to capture in real time.

Educational Problem

The main problem in this setting is the lack of effective, scalable assessment methods for maker learning. Maker education emphasizes creativity, iteration, problem-solving, and design reasoning, yet current assessment practices do not capture these skills well.

Teachers report several specific issues:

- Limited visibility into student thinking. Students make many decisions during hands-on work, but teachers cannot observe every step.
- Uneven feedback. Some students receive detailed guidance, while others receive minimal support because the teacher is assisting another group.
- Inconsistent assessment. Teachers rely on subjective judgments of effort, creativity, or "quality," resulting in variation across classrooms.
- Final products overshadow the process. Students who struggle during the process but produce a neat final product may appear successful, while those who iterate deeply may not receive credit for their learning.
- Multilingual learners are overlooked. Students who are quiet or still developing academic English may not articulate their reasoning, making their learning difficult to assess.

These issues are well-documented in maker-education research. Blikstein (2013) found that teachers struggle to assess process-based learning because they cannot track the many micro-decisions students make during hands-on work. Halverson and Sheridan (2014) similarly noted that maker learning produces rich but hard-to-capture evidence of thinking.

If this problem persists, assessments will influence instructional decisions, student motivation, and the perceived value of maker learning. When assessments fail to capture the skills that maker education aims to develop, teachers cannot provide targeted support, and students do not receive feedback that helps them improve.

This gap represents a genuine educational problem, not merely a logistical one. When assessment does not reflect what students actually learn, it distorts instructional decisions, limits equitable access to feedback, and undermines the goals of maker education. Students who iterate deeply but produce imperfect final products may be misjudged, while those who struggle silently, particularly multilingual learners, may never be identified as needing support. As Black and Wiliam (1998) established, assessment and learning are inseparable; when assessment breaks down, so does the opportunity to improve instruction and student outcomes.

Existing and New Solutions

Existing Approaches

Schools currently use several tools to assess maker learning:

- Rubrics for creativity, engineering design, or collaboration
- Reflection journals where students describe their process
- Peer feedback during design reviews
- Digital portfolios that store photos, videos, and artifacts

These tools help document learning, but they have limitations. Rubrics often emphasize final products over the process. Journals are time-consuming to review. Peer feedback varies in quality. Digital portfolios capture artifacts but not the reasoning behind them. Teachers still struggle to track how students revise their ideas or respond to challenges.

New Direction: AI-Supported Personalization

Recent research in learning analytics and Artificial Intelligence (AI)-supported instruction provides concrete evidence that AI can help teachers monitor student progress on open-ended tasks. Several studies demonstrate this potential:

- Blikstein and Worsley (2016) showed that multimodal learning analytics can offer new insights into students' learning trajectories in complex, open-ended environments such as makerspaces.
- Holstein et al. (2019) demonstrated that AI-supported teacher dashboards can deliver real-time insights into student learning during complex tasks.
- Schneider and Blikstein (2015) showed that multimodal learning analytics can detect meaningful patterns in student interactions with tangible learning environments, informing personalized feedback during hands-on tasks.
- VanLehn (2011) showed that intelligent tutoring systems can provide individualized guidance during problem-solving tasks.

These studies provide evidence that AI can:

- detect patterns in student actions.
- identify common misconceptions.
- track iteration steps.
- generate personalized prompts.

- provide consistent feedback.

This direction addresses a clear gap: teachers need a practical way to assess process-based learning without interrupting hands-on work.

Theoretical Framework

Several learning theories support the design of an AI-supported assessment system.

Constructionism

Constructionism states that students learn by creating meaningful artifacts (Papert, 1980). Because learning occurs during the making process, assessment must capture how students plan, test, and revise their designs. AI can help document these steps.

Formative Assessment Theory

Effective formative assessment provides timely, actionable feedback that helps students adjust their work (Black & Wiliam, 1998). AI can deliver immediate prompts during maker tasks, something teachers cannot always provide for every student.

Self-Determination Theory (SDT)

Self-Determination Theory emphasizes competence, autonomy, and relatedness (Deci & Ryan, 2000). Personalized feedback can help students feel supported and capable, especially for those who hesitate to ask for help.

Cognitive Apprenticeship

Cognitive apprenticeship emphasizes the importance of modeling expert thinking and providing scaffolds (Collins et al., 1991). AI can serve as an additional scaffold by identifying when students need guidance and offering suggestions.

Ontological Innovation

Ontological innovation explains how new concepts emerge when existing categories fail to fully describe learning (diSessa & Cobb, 2004). Maker learning often reveals patterns, such as persistence, depth of iteration, or design flexibility, that traditional assessments miss. AI may help identify and define these new constructs.

How These Theories Shape the Intervention

These theories guide the design of the AI-supported assessment system in concrete ways:

- Constructionism supports the making process rather than only the final product.
- Formative assessment supports the provision of real-time feedback during hands-on work.
- Self-Determination Theory supports the design of feedback that builds confidence and encourages persistence.
- Cognitive apprenticeship supports the use of AI to model expert reasoning and scaffold next steps.
- Ontological innovation supports the identification of new assessment constructs that better represent maker learning.

Together, these theories provide a foundation for designing an intervention that is practical, equitable, and aligned with the goals of maker education. These theoretical conjectures lead to the following research questions that will guide the design and evaluation of the proposed intervention:

- How can an AI-supported assessment system be designed to capture process-based learning behaviors, such as iteration, troubleshooting, and design reasoning, in K–12 maker education classrooms?
- To what extent does AI-generated personalized feedback improve the consistency and equity of formative assessment for diverse learners in makerspace environments?
- How do teachers perceive and use AI-generated insights into student learning to inform their instructional decisions during maker activities?

References

Black, P., & Wiliam, D. (1998). Assessment and classroom learning. *Assessment in Education: Principles, Policy & Practice, 5*(1), 7–74. https://doi.org/10.1080/0969595980050102

Blikstein, P. (2013). Digital fabrication and 'making' in education: The democratization of invention. In J. Walter-Herrmann & C. Büching (Eds.), *FabLabs: Of machines, makers and inventors* (pp. 1–21). Transcript Publishers. https://doi.org/10.14361/TRANSCRIPT.9783839423820.203

Blikstein, P., & Worsley, M. (2016). Multimodal learning analytics and education data mining: Using computational technologies to measure complex learning tasks. *Journal of Learning Analytics, 3*(2), 220–238. https://doi.org/10.18608/jla.2016.32.11

Collins, A., Brown, J. S., & Holum, A. (1991). Cognitive apprenticeship: Making thinking visible. *American Educator, 15*(3), 6–11.

Deci, E. L., & Ryan, R. M. (2000). The "what" and "why" of goal pursuits: Human needs and the self-determination of behavior. *Psychological Inquiry, 11*(4), 227–268. https://doi.org/10.1207/S15327965PLI1104_01

diSessa, A. A., & Cobb, P. (2004). Ontological innovation and the role of theory in design experiments. *Journal of the Learning Sciences, 13*(1), 77–103. https://doi.org/10.1207/s15327809jls1301_4

Halverson, E. R., & Sheridan, K. (2014). The maker movement in education. *Harvard Educational Review, 84*(4), 495–504. https://doi.org/10.17763/haer.84.4.34j1g68140382063

Holstein, K., McLaren, B. M., & Aleven, V. (2019). Co-designing a real-time classroom orchestration tool to support teacher–AI complementarity. *Journal of Learning Analytics, 6*(2), 27–52. https://doi.org/10.18608/jla.2019.62.3

Kafai, Y. B., & Peppler, K. A. (2011). Youth, technology, and DIY: Developing participatory competencies in creative media production. *Review of Research in Education, 35*(1), 89–119. https://doi.org/10.3102/0091732X10383211

Papert, S. (1980). *Mindstorms: Children, computers, and powerful ideas*. Basic Books.

Schneider, B., Blikstein, P., & Pea, R. (2021). Multimodal learning analytics in makerspaces. *Journal of Learning Analytics, 8*(1), 1–15. https://doi.org/10.5281/zenodo.3554729

VanLehn, K. (2011). The relative effectiveness of human tutoring, intelligent tutoring systems, and other tutoring systems. *Educational Psychologist, 46*(4), 197–221. https://doi.org/10.1080/00461520.2011.611369